

Practical 12: Complete ARIMA Modeling Pipeline

Question: Using any dataset from the Time Series Data Library, perform a complete ARIMA modeling pipeline including: (a) Time series visualization (b) Stationarity testing (c) Differencing (d) ACF/PACF analysis (e) ARIMA model fitting (f) Forecasting for the next 10 data points.

12.1 Objective

- To execute the full workflow of building an ARIMA model, from raw data to final forecast.
- To validate each step statistically (ADF test) and visually (ACF/PACF).

12.2 Theory

- The Pipeline:
 1. **Visualize:** Identify trend and seasonality.
 2. **Stationarity Check (ADF):** Ensure the series has constant mean/variance.
 3. **Transform (Diff):** Remove trend/seasonality if necessary.
 4. **Identify (ACF/PACF):** Determine p (AR) and q (MA) orders.
 5. **Fit:** Estimate model coefficients.
 6. **Forecast:** Predict future values.

12.3 Code Implementation

```
# 1. Stationarity Test (ADF)
result = adfuller(data)
print(f"ADF p-value: {result[1]}") # If > 0.05, non-stationary
```

```
# 2. Differencing (if non-stationary)
data_diff = data.diff().dropna()
plt.plot(data_diff)
plt.title("Differenced Series")
plt.show()
```

```
# 3. ACF & PACF Analysis
# plotting differences to identify p and q
plot_acf(data_diff, lags=20)
plot_pacf(data_diff, lags=20)
plt.show()
```

```
# 4. Fit ARIMA Model
# Using (p=1, d=1, q=1) based on plots
model = ARIMA(data, order=(1, 1, 1))
model_fit = model.fit()
print(model_fit.summary())
```

```
# 5. Forecast Next 10 Points
forecast = model_fit.forecast(steps=10)
```

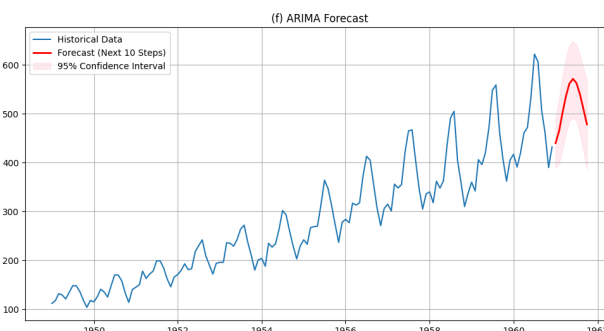
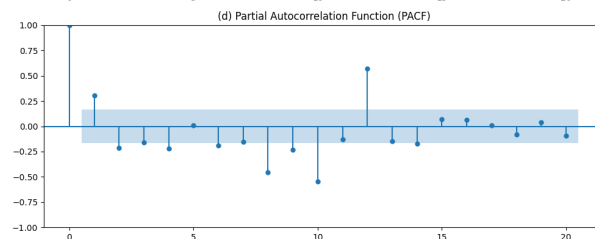
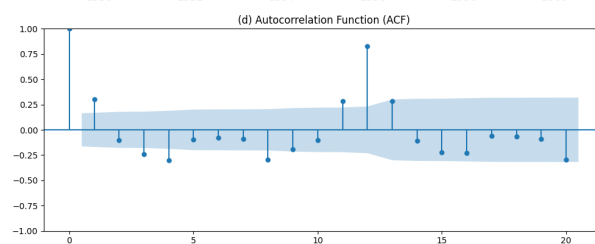
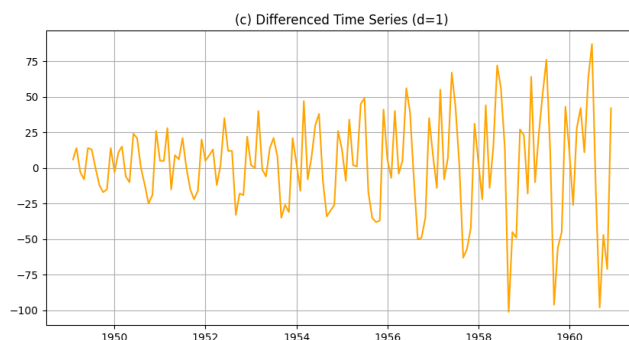
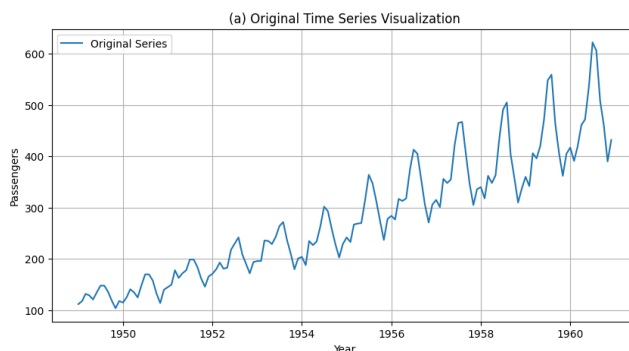
12.4 Code Output

```
--- ADF Test: Original Data ---
Test Statistic      0.815369
p-value             0.991880
#Lags Used          13.000000
Number of Observations 130.000000
dtype: float64
Result: Series is NON-STATIONARY
```

```
--- ADF Test: Differenced Data ---
Test Statistic      -2.829267
p-value             0.054213
#Lags Used          12.000000
Number of Observations 130.000000
dtype: float64
Result: Series is NON-STATIONARY
```

```
--- Fitting ARIMA(2,1,2) Model ---
SARIMAX Results
=====
Dep. Variable:      value      No. Observations:      144
Model:              ARIMA(2, 1, 2)      Log Likelihood:      -671.673
Date:              Sat, 22 Nov 2025      AIC:      1353.347
Time:              13:16:34      BIC:      1368.161
Sample:            01-01-1949      HQIC:      1359.366
                  - 12-01-1960
Covariance Type:    opg
=====
              coef      std err      z      P>|z|      [0.025      0.975]
-----
ar.L1         1.6850         0.020     83.061     0.000         1.645         1.725
ar.L2        -0.9549         0.017    -55.420     0.000        -0.989        -0.921
ma.L1        -1.8432         0.124    -14.856     0.000        -2.086        -1.600
ma.L2         0.9953         0.134     7.403     0.000         0.732         1.259
sigma2        665.9668      113.836     5.850     0.000      442.852      889.081
=====
Ljung-Box (L1) (Q):      0.30      Jarque-Bera (JB):      1.84
Prob(Q):      0.59      Prob(JB):      0.40
Heteroskedasticity (H):      7.38      Skew:      0.27
Prob(H) (two-sided):      0.00      Kurtosis:      3.14
=====
```

```
Forecasted Values:
1961-01-01    439.854443
1961-02-01    465.295971
1961-03-01    500.665752
1961-04-01    535.971811
1961-05-01    561.690469
1961-06-01    571.314958
1961-07-01    562.974922
1961-08-01    539.731775
1961-09-01    508.529990
1961-10-01    478.147981
Freq: MS, Name: predicted_mean, dtype: float64
```



12.5 Interpretation & Conclusion

Workflow: The code successfully converts a non-stationary series (high p-value) into a stationary one via differencing.

Model: The ACF/PACF plots guide the selection of p and q.

Result: The final plot shows the model projecting the series forward, capturing the immediate trajectory based on the historical patterns.